

Predicting Infections during hospitalization among postsurgical patients

A multimodal ML + NLP framework for early infection risk

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Hospital-acquired infection: the clinical problem

2 - 5 %

of operations develop a SSI^[1]

3.5 - 10 B\$

excess annual cost from SSIs
in the US alone ^[2]

0

national SSI surveillance
programs in Israel ^[3]

Today- detection of infection is usually confirmed only once it is already apparent.

^[1] World Health Organization. *Global Guidelines for the Prevention of Surgical Site Infection*, 2nd ed. Geneva: WHO, 2018.

^[2] Chakraborty A, et al. "Development and evaluation of machine learning models for the identification of surgical site infection in electronic health records." *Surgical Infections* 26, no. 7 (2025): 474.

^[3] Najjar Debbiny R, et al. "Healthcare-associated infection prevention and control practices in Israel: results of a national survey." *BMC Infectious Diseases* 22 (2022): 739.

The research gap

01

**Structured models often lack
external validation^{[4][5]}**

02

**Text adds value - but
representation matters^{[6][7]}**

03

**Hebrew multimodal SSI prediction
remains unexplored^[8]**

Research question:

To what extent can a multimodal machine learning framework, combining structured tabular EHR data with unstructured narrative clinical text, predict the risk of postoperative infection at the end of the surgical procedure among surgical patients at Assuta Ramat HaHayal, compared with a single-modality tabular model?

[4] Ndjongo LCM, et al. "Evaluating predictive performance and generalizability of traditional and artificial intelligence models in predicting surgical site infections post-spinal surgery: a systematic review." Spine Journal 26, no. 2 (2026): 280–291.

[5] Cui Y, Shi X, Wang Q, et al. "Artificial intelligence-based prediction model for surgical site infection in metastatic spinal disease: a multicenter development and validation study." International Journal of Surgery (2025).

[6] Bonde A, et al. "Assessing the utility of deep neural networks in detecting superficial surgical site infections from free text health record data." Frontiers in Digital Health 5 (2024): 1249835.

[7] Verberk JDM, et al. "The augmented value of using clinical notes in semi-automated surveillance of deep surgical site infections after colorectal surgery." Antimicrobial Resistance & Infection Control 12 (2023): 117.

[8] Saban M, et al. "Identifying diabetes related-complications in a real-world free-text electronic medical records in Hebrew using natural language processing techniques." Journal of Diabetes Science and Technology 19, no. 4 (2025): 99–1007.

Project goals

- 1 Develop a data-driven ML model to predict the risk of SSI during hospitalization.
- 2 Enable early identification of elevated-risk patients.
- 3 Identify key variables associated with infection risk.
- 4 Evaluate the contribution of these variables to prediction performance.
- 5 Compare the performance of different predictive models and select the most accurate and reliable approach.

Final scope: postoperative SSI prediction at the end of surgery.

The data: Assuta Ramat HaHayal EHR

35,925 surgical events

141 · 0.39% Positive SSI

244 → 97 Columns after cleaning

Figure 2- the outcome is very rare

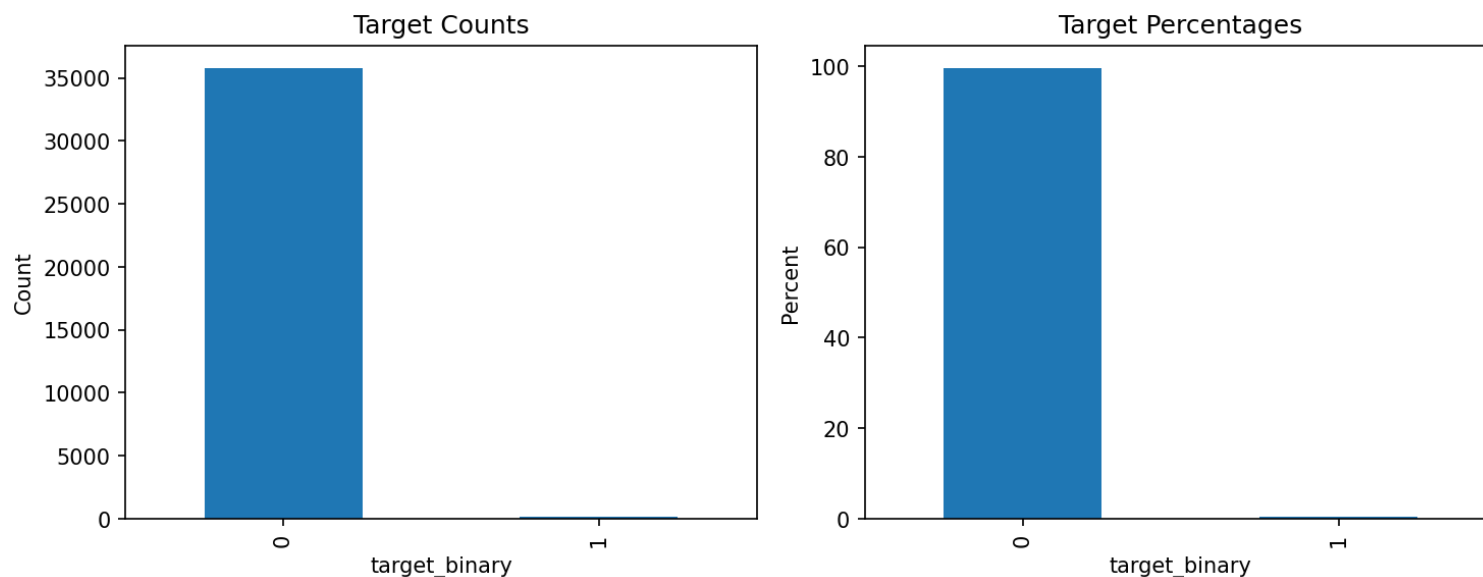
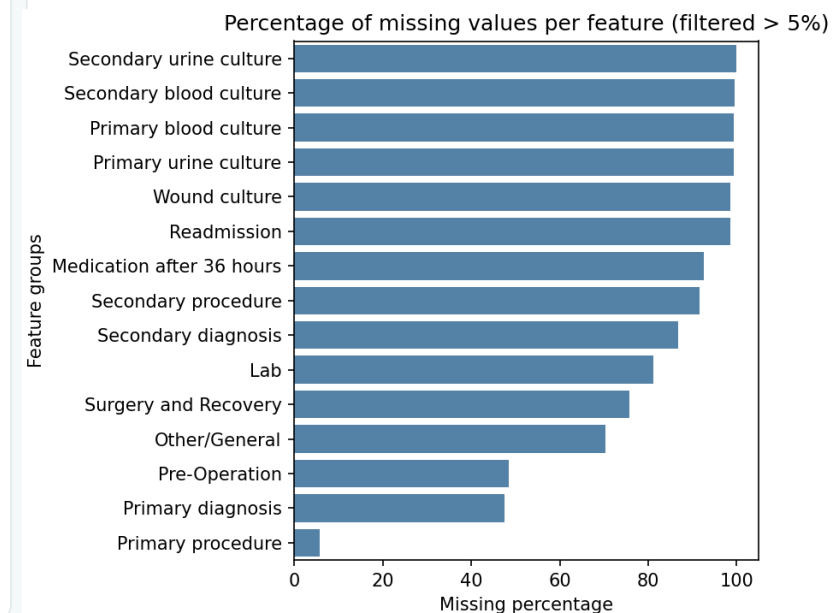
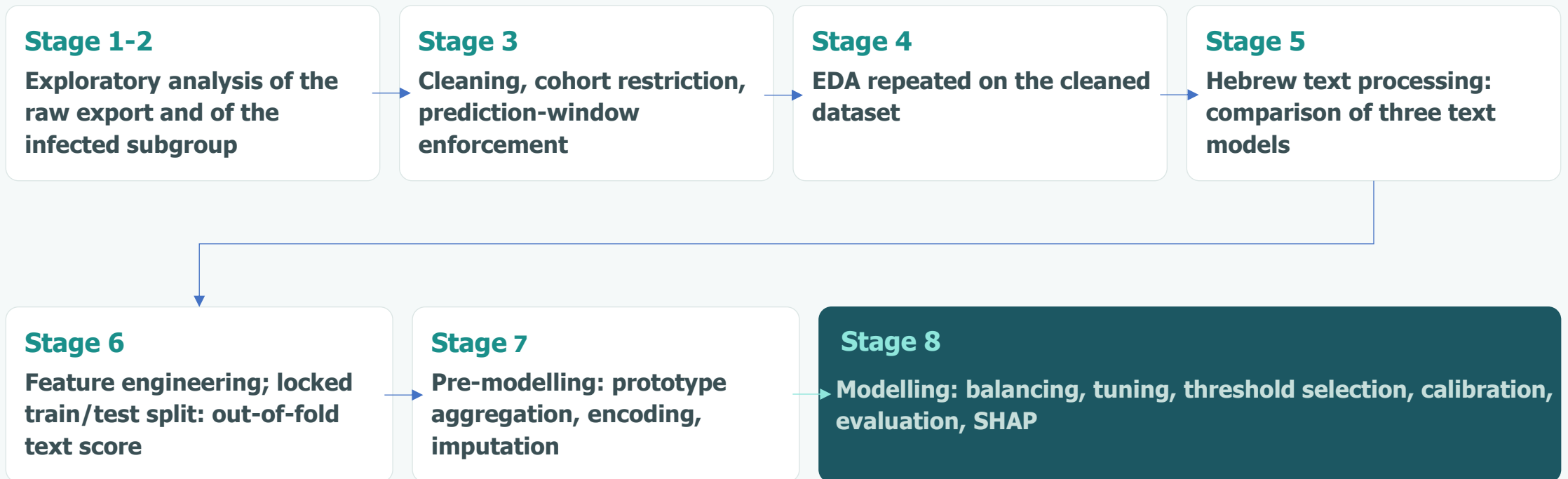


Figure 3- missingness by feature group



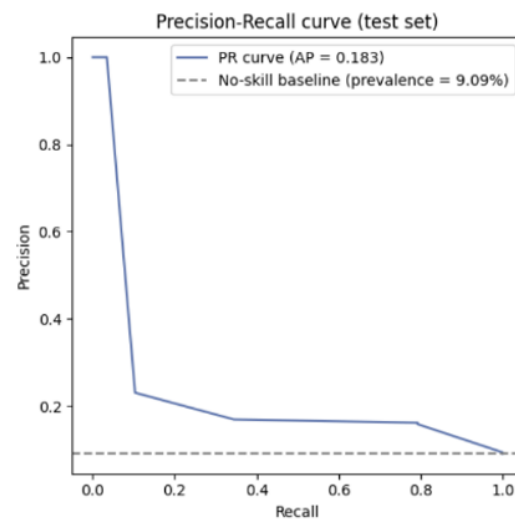
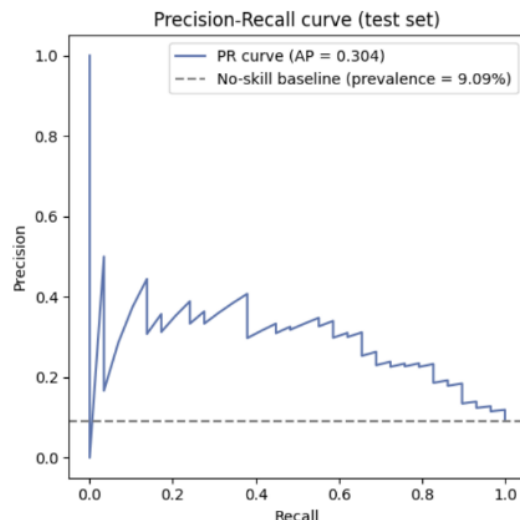
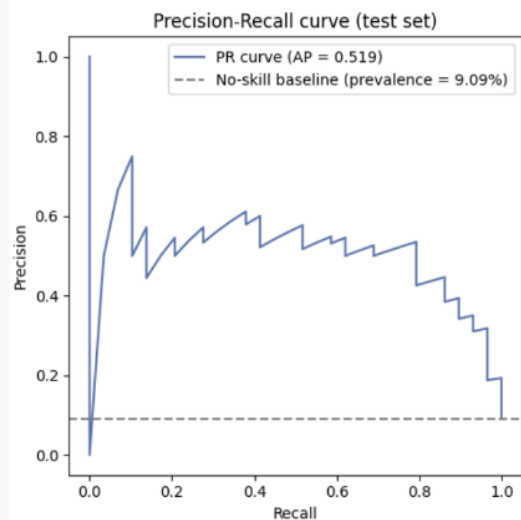
From raw export to a risk model



Comparing the text models

Method	ROC-AUC	PR-AUC (AP)	Precision	Recall
TF-IDF + logistic regression	0.933	0.519	0.577	0.517
TF-IDF + XGBoost	0.821	0.304	0.286	0.069
Llama 3.3:70b, prompted	0.701	0.183	0.169	0.345

Precision–recall by representation



Evaluated on a rebalanced 319-note split (29 positive), built to rank the three representations

The simplest model won decisively over a 70-billion-parameter LLM

SECOND COMPARISON

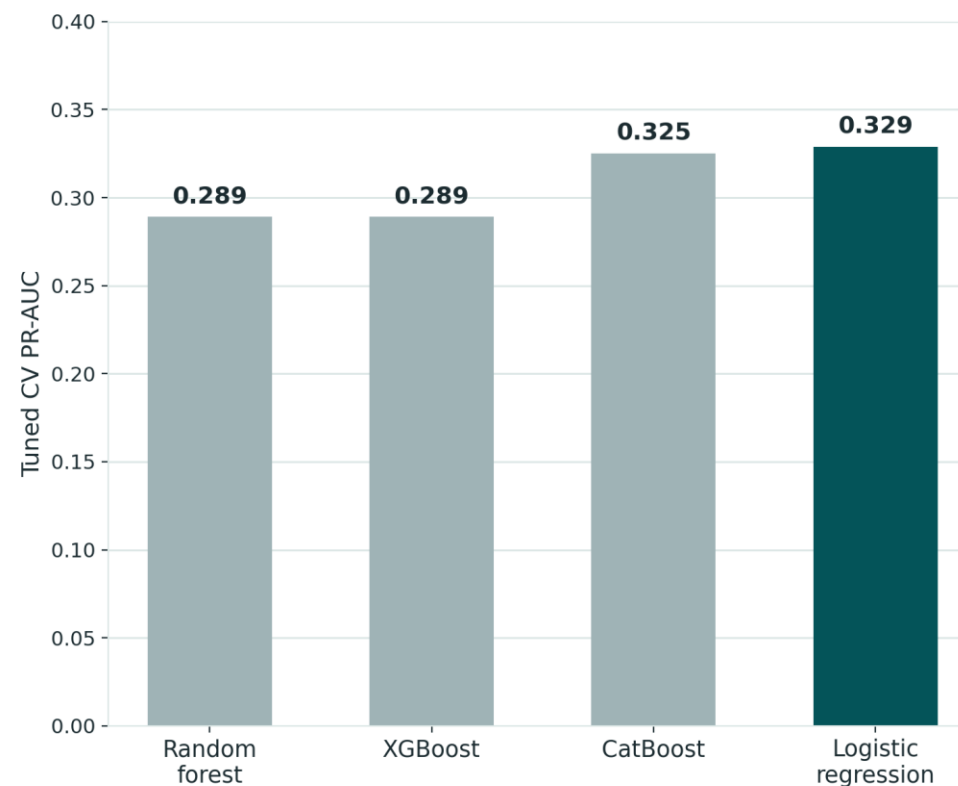
Comparing the tabular models

Model	Tuned CV PR-AUC	Tuning runtime (s)
Logistic regression - locked	0.329	11
CatBoost	0.325	1,547
XGBoost	0.289	31
Random forest	0.289	86

Takeaway

Logistic regression and CatBoost were a near-tie on CV PR-AUC. The tie was resolved for the cheaper, more interpretable SMOTE-balanced linear model.

Tuned CV PR-AUC by model



Final model performance

88.1% recall- 37/42 caught

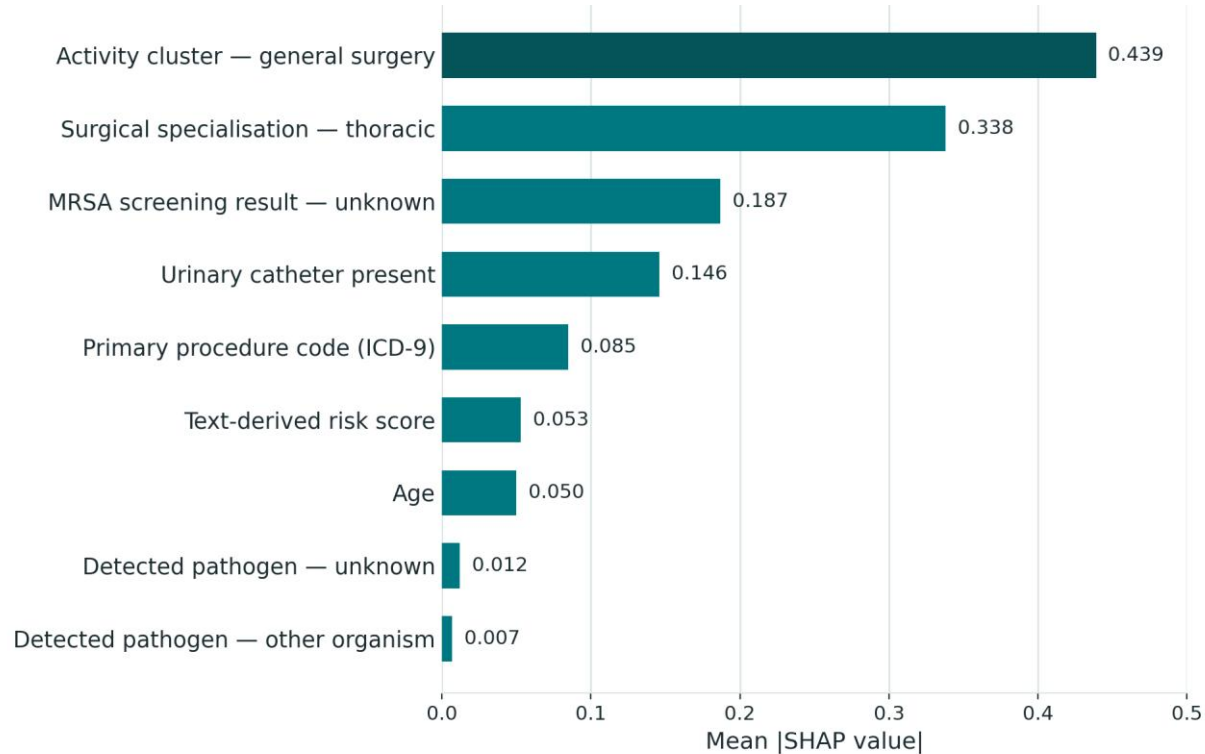
4.7% accuracy

0.183 PR-AUC

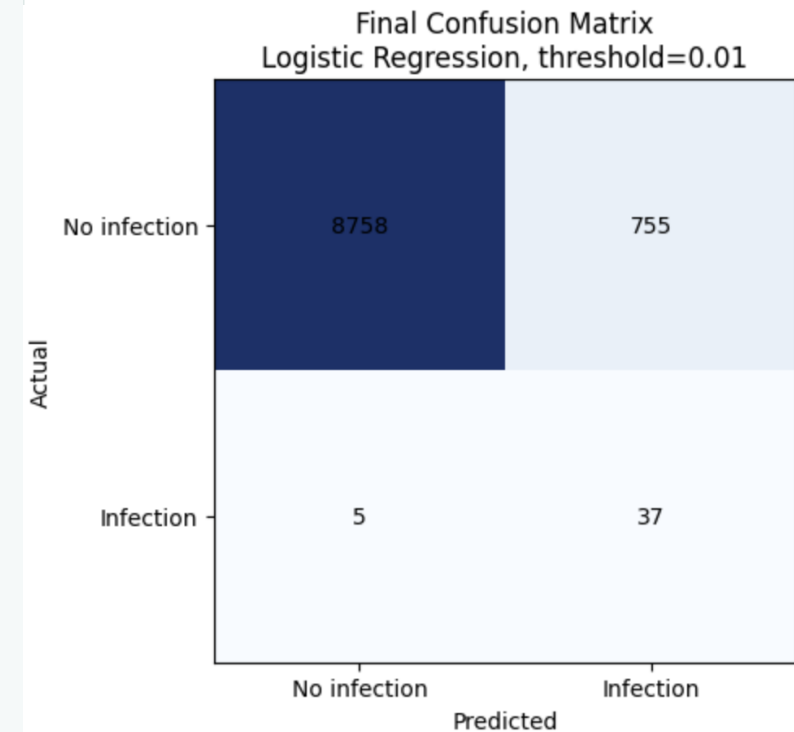
21.4 NNS

0.0039 Brier score- calibration

What drives the score (SHAP)



Confusion matrix - 9,555 events, 42 infections



Clinical workload: 82.9 alerts per 1,000 procedures.

Conclusions

- 1 Hebrew clinical narrative carries genuine infection**

- 2 High discrimination does not mean high precision**

- 3 The prediction point is a clinical decision, not a technical parameter**

- 4 A single scalar may be too narrow a channel for multimodal integration**

- 5 Leakage control has to be designed into the pipeline**

Objectives: 4 achieved / 1 partially achieved- early identification demonstrated retrospectively, not prospectively.

Limitations

Statistical power

42 positive cases in the test set.

A compressed feature set

266 features were reduced to 20 before modelling.

Pre-operative by clinical definition

No patient-level timestamp

Possible cohort artefact

Four zero-infection activity clusters were excluded

Limited post-discharge follow-up

Infections outside Assuta missed

Performance varies by subgroup

Large subgroup differences

Directions for further work

Richer text integration

Beyond one scalar score

Hebrew Transformer

Domain-adapted, GPU-trained

Subgroup-specific thresholds

Especially thoracic surgery

Prospective evaluation

On newly admitted patients

Dynamic risk model

Continuously updating

Hospital dashboard

Surface score in workflow

Thank you!

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